

Review

Chemicals mediate the interaction between
plant-associated fungi and insectsYang Xu¹ , Lei Xu² , and Hong He^{1*}¹College of Forestry, Northwest A&F University, Yangling 712100, Shaanxi, China²College of Life Sciences, Northwest A&F University, Yangling 712100, Shaanxi, China

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Abstract Plants, insects, and fungi have successfully colonized almost all terrestrial ecosystems, and their interactions have been the subject of numerous studies in recent decades. Plant-associated fungi include endophytic, arbuscular mycorrhizal, ambrosia, saprotrophic, pathogenic, and floral fungi. These fungi interact with insects through various mechanisms, including the modification of plant nutritional quality and degradation of plant defensive allelochemicals that are toxic to insects. Additionally, certain fungi assist plants in defending against insect attacks. Correspondingly, insects have evolved sophisticated nervous, digestive, and muscular systems that assist them in recognizing, preying on, and dispersing plant-associated fungi; these organ systems allow insects to detect and respond to various chemical signatures in the environment. Insects can be nourished, attracted, repelled, poisoned, and killed by chemical molecules produced by plant-associated fungi, which could be beneficial or detrimental to plants. This review summarizes the functions of different chemicals from the perspective of plant–fungus–insect interactions and discusses the challenges and future perspectives in this chemical ecology research field.

Key words: attraction, chemical, insect, plant-associated fungi, repellent.

1 Introduction

Plants, insects, and fungi are essential components of an ecosystem; their ecological niches are interconnected in trophic networks and intricate relationships, including predatory, saprophytic, mutualistic, competitive, parasitic, and pathogenic interactions (Gange & Brown, 2002; Lilleskov & Bruns, 2005; Moonjely et al., 2016; Ljunggren et al., 2019). Most fungi are highly nutritious but nonmotile. Multiple studies proved that some fungi are more nutritious than plant tissue resources for insects (Bentz & Six, 2006; Raman & Suryanarayanan, 2017). In most cases, plant-associated fungal species assist their plant hosts in influencing insect behavior, such as by attracting insect pollinators or deterring herbivorous insects (Guerrieri et al., 2004; Barker et al., 2015; Becher et al., 2018).

Plants and fungi are nonmotile eukaryotes that possess diverse metabolic pathways; various metabolites are used to interact with insects. Meanwhile, insects are mobile eukaryotes that use their chemosensory systems to rapidly detect and distinguish different chemical signatures in their environment. Multiple studies have shown that the interactions between plant-associated fungi and insects are characterized by significant alterations in the chemicals produced by the nonmotile organisms (Doll et al., 2013; Barker et al., 2015; Shrivastava et al., 2015; Xu et al., 2019). Specifically, fungi or host plants produce molecules that

function either from a distance or through contact; these molecules can influence insect behavior by serving as attractants, repellents, antifeedants, toxins, and even insecticides (Boucias et al., 2012).

The interactions between plants, fungi, and insects are common in natural ecosystems. However, current research primarily focuses on individual interactions between plants and fungi or between fungi and insects. This limited approach fails to consider the collective importance of these interactions in ecological settings (Drilling & Dettner, 2009; Zhao et al., 2015; Wu & Duncan, 2020). To provide insights into plant–fungus–insect interactions, this review was prepared to consolidate and classify the plant responses, fungal colonization, and insect behavior that have been overlooked in previous studies.

The chemical compounds in plants and fungi are involved in nutrient circulation (Douglas, 1994; Bleiker & Six, 2007), chemical defense, and influencing insect behavior. In this review, we enumerated the chemicals that mediate the communication between plant-associated fungi and insects. We also emphasized that the chemicals produced by host plants associated with fungi affect insect behavior. In addition, we summarized the possible ecological significance of these chemicals. We hope that this review will stimulate interest in chemically mediated insect–fungi interactions, along with potential biotechnological applications of these signals.

2 Fungi Mediate Trophic Interactions Between Plants and Insects

Plants, fungi, and insects are intricately related at the nutritional level (Fig. 1). Although plants directly provide some nutrition to their colonized fungi and phytophagous insects, they are not considered optimal food sources for some insects because plant biomass consists of indigestible compounds (cellulose and lignin) and allelochemicals (Douglas, 1994; Li et al., 2021). Fungi facilitate the flow of energy in different ecosystems. They support insect survival by converting high-carbon cellulose substrates into balanced nutritional sources (Engel & Moran, 2013). Moreover, plant-associated fungi provide habitat and diverse nutrition for insects. Specifically, some fungi act as “early colonizers” and accumulate valuable free nutrients that can enrich free amino acids, nonstructural carbohydrates, organic nitrogen-containing compounds, vitamins, and lipids in plant or fungal tissues (Jones, 1963; Ayres et al., 2000; Bleiker & Six, 2007; Watkinson, 2016). Such enrichment of essential biomolecules balances the C: N ratio in plant or fungal tissues, resulting in improved herbivore performance. Some fungal symbionts are wood-decaying organisms, which produce cellulolytic enzymes that facilitate wood digestion by insects (Henriques et al., 2006). For example, the ambrosia beetle *Mycangia* carries and cultures filamentous fungi and yeast in their galleries to acquire nutrition (Mueller & Gerardo, 2002; Li

et al., 2015; Yun et al., 2015; Hulcr & Stelinski, 2017). In addition, the symbiont fungus *Termitomyces*, farmed by termites, can provide complete nutrition for its ant hosts (Schalk et al., 2021).

Insects utilize fungal enzymes to detoxify plant defense chemicals (Boddy & Jones, 2008). Various fungi associated with termites, leaf-cutting ants, beetles, and wasps have been reported to detoxify plant allelochemicals, including tannins, quinones, phenols, alkaloids, terpenes, and fatty acids (Dowd, 1992). It has been revealed that the cytochromes of bark beetle symbiotic fungi and gut-associated fungi assist insect hosts in metabolizing plant defense chemicals, such as terpenoids and phenolics (Wadke et al., 2016; Dai et al., 2022). In addition, the leaf-cutting ant *Acromyrmex echinator* feeds on their cultivated *Leucocoprinus gongylophorus*, which produces laccase—an enzyme that passes through the digestive tract of ants and is released onto a plant surface, aiding the ant in degrading plant defense chemicals, such as tannins and flavonoids (De Fine Licht et al., 2013).

Interestingly, some fungi are both mutualistic with plants and pathogenic to insects and termed endophytic insect pathogenic fungi (EIPF); examples include *Beauveria* sp. and *Metarhizium* sp. (Hu & Bidochka, 2021). In this symbiotic interaction between plants and EIPF, plants benefit by acquiring nitrogen derived from insects, while fungi benefit by obtaining photosynthates transported by

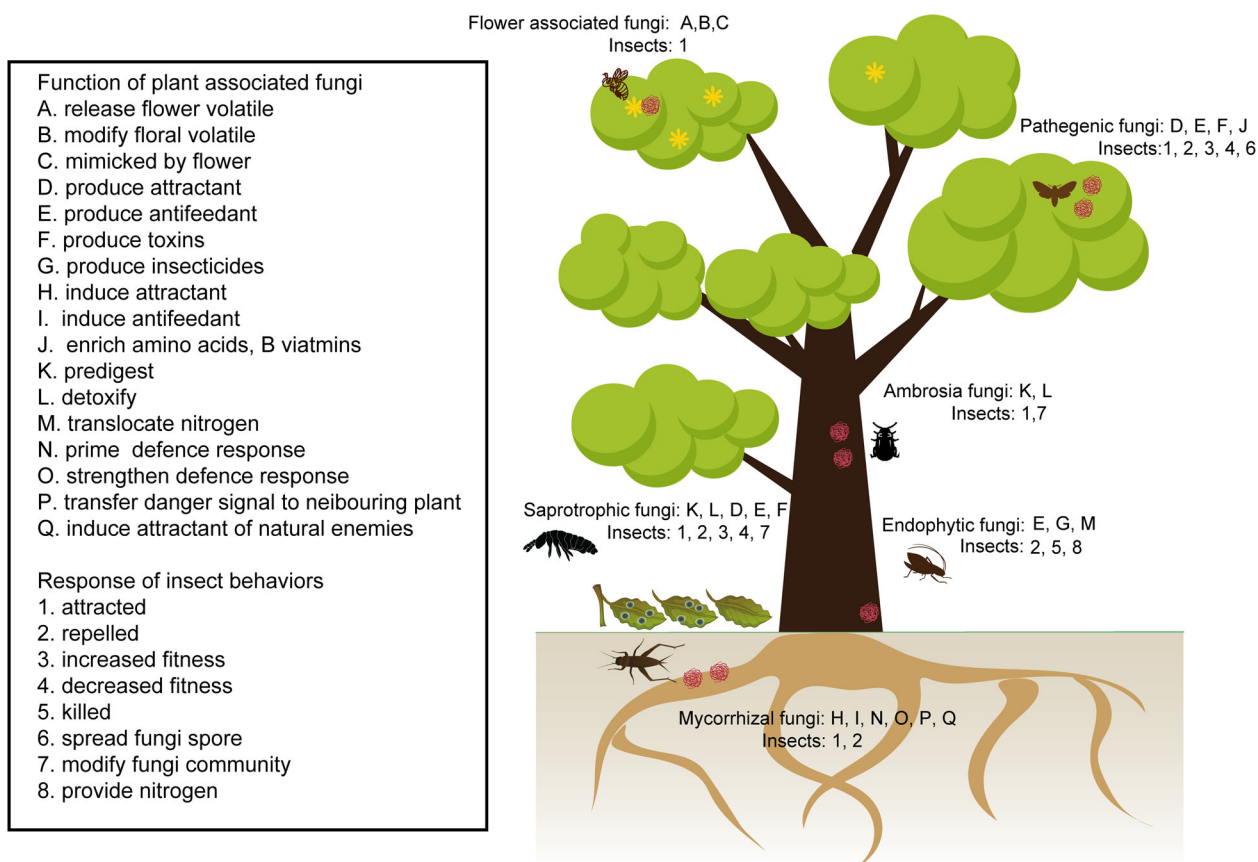


Fig. 1. Schematic classification of chemical-related strategies used by plant-associated fungi to interact with insects.

plants. Such a phenomenon was proven by an experiment utilizing radioactive isotopes ^{15}N and ^{13}C , which indicate nutrient transfer across fungi, plants, and insects (Behie et al., 2012).

3 Fungi-modulated Chemicals Change Insect Behavior

Aside from being closely connected at the nutritional level, plant-associated fungi and insects also interact through other levels of relationships, such as pollination and dispersal of fungal spores. Chemical compounds play a central role in these interactions (Table 1). Meanwhile, insects evolved sophisticated chemosensory systems to interact with plants and fungi (Halitschke et al., 2001; Leal, 2013; Schaeffer et al., 2019). Generally, the chemicals that mediate the interactions between plant-associated fungi and insects are classified into four categories.

First, plant-associated fungi produce compounds that deter or attract insect herbivores depending on their constitutive chemical production (Table 1a). Due to their shared evolutionary history with animals, a majority of secondary metabolites produced by plant-associated fungi are essential for modulating insect behavior (Rohlfes & Churchill, 2011; Franco et al., 2021). These metabolites perform various ecological functions: either attracting pollinators, herbivores, and predators or deterring pests by producing toxins and pungent or bitter compounds in the fruiting body (Daisy et al., 2002; Faria & Wraight, 2007; Spiteller, 2008).

Second, fungal colonization alters the metabolites of a host plant; modification of defense compounds or attractive volatiles further modulates herbivore behavior (Bush et al., 1993; Kaiser, 2006; Mayer et al., 2008) (Table 1b). Fungi colonization primes the defense responses of host plants, resulting in faster and stronger responses to herbivory. For example, endophytic fungi and mycorrhizal species increase the terpenoid and pyrrolizidine alkaloid levels in the host plants, bolstering their defense against herbivorous insects (Bush et al., 1993; Barker et al., 2015; Shrivastava et al., 2015; Popay & Rowan, 2017; Hill et al., 2018). In some instances, pathogenic fungi could induce production of mannitol, total nitrogen, essential amino acids, and B vitamins in black poplar, making the host plants more attractive to pests (Eberl et al., 2020). Notably, the influence of plant-associated fungi on herbivores varies and is largely dependent on insect foraging patterns (generalists, specialists, chewers, and suckers) and their enemies, such as parasitoids. For instance, arbuscular mycorrhizal fungi (AMF) colonization negatively affects generalist below-ground herbivores, such as parasitic nematodes and beetle larvae (Gange & West, 1994; Rodríguez-Echeverría et al., 2009). However, their colonization positively affects the root-feeding larvae of specialist weevils (Currie et al., 2011).

Third, insect behavior modulates the defense compounds in plant-associated fungi. Certain chemicals in these fungi are produced or induced only when they are attacked by insect predators (Table 1c). The chewing of insects on plant-associated fungi physically damages the fungal tissue, triggering a cascade of reactions that lead to the production

of aurofusarin, viomellein, and xanthomegnin, which repel various predators (Xu et al., 2019). The production of these chemicals is an energy-demanding process; thus, injured organisms should produce potential chemical weapons to cope with stressful ecological conditions. Insect activity affects the intrinsic palatability and toxicity of fungi, eventually influencing insect-feeding activities in plants. Other studies have convincingly demonstrated that fungi chemically respond to insect predator attacks by releasing repellents and toxicants for insect predators (Rohlfes et al., 2007; Spiteller, 2008; Rohlfes & Churchill, 2011; Doll et al., 2013; Xu et al., 2019).

Finally, insect behavior induces chemical changes in plants with the involvement of fungi; such chemical changes modulate the behavior of subsequent insect visitors (Table 1d). In the presence of AMF, aphid predation induces host plants to produce (Z)-3-hexenyl acetate, naphthalene, and (R)-germacrene D; these chemicals attract parasitoids (the natural enemies of aphids) (Babikova et al., 2014a). Moreover, through common mycorrhizal connections, the corresponding signal is transferred to neighboring plants; consequently, they produce more defense compounds that are repellent to aphids but attractive to parasitoids (Babikova et al., 2014b; Song et al., 2019). In addition, insect feeding on plant material either inhibits or stimulates the biomass of plant-associated fungi, such as mycorrhizal and endophytic fungi and such effects further influence the defense abilities of plants (Babikova et al., 2014a; Wilkinson et al., 2019). Although some of the key chemicals involved in these processes have not been identified, their functions have not been well studied.

The chemicals that influence insect behavior and are emitted by plants (Table 1b, 1d) and their associated fungi (Table 1a, c) have been reported. Volatile organic compounds are airborne metabolites, such as pheromones, allomones, and kairomones, that mediate chemical communication at a distance (Kandasamy et al., 2019). In addition, nonvolatile compounds play essential contact-dependent roles in exerting toxic or antifeedant effects. Notably, these chemical signals have different effects on feeding insects with varying degrees of specialization in herbivores (Gange et al., 2012). Although the effects of the chemicals constitutively produced by plant-associated fungi have been well studied, the functions of chemicals produced by host organisms under the influence of other organisms have not been thoroughly identified. Hence, the roles of these chemicals should be determined to gain a comprehensive understanding of the communication between plants, insects, and fungi.

4 Ecological Significance of Plant-associated Fungi that Interact with Insects

4.1 Influence of plant organic matter decay and nutrient cycling

The interaction between plant-associated fungi and insects helps maintain ecosystem balance and contributes to dead plant biomass decomposition and nutrient cycling in terrestrial habitats. Nearly 90% of terrestrial plant production

Table 1 Characterization of chemical compounds that mediate interactions between plant-associated fungi and insects

Fungi	Insect	Relationship	Chemical	Function	Research
<i>a. Produced by plant-associated fungi</i>					
<i>Fusarium solani</i>	<i>Drosophila melanogaster</i>	Saprophytic	1-Octen-3-ol	Attract	Wu & Duncan (2020)
<i>Trametes gibbosa</i>	<i>Cis boleti</i>	Saprophytic	1-Octen-3-ol	Attract	Thakeow et al. (2008)
<i>T. versicolor</i>	<i>Dacne bipustulata</i>	Saprophytic	δ -Cadinene, β -guaiene, isolekene, γ -patchoulene	Attract	Drilling & Dettner (2009)
	<i>Tritoma bipustulata</i>				
	<i>Sulcaxis affinis</i>				
	<i>Diaperis boleti</i>				
<i>Endoconidiophora polonica</i>	<i>Ips typographus</i>	Symbiotic	Bicyclic ketals, exo-brevicommin, endo-brevicommin, trans-conophthorin, exo-1,3-dimethyl-2,9-dioxabicyclo[3.3.1]nonane, endo-1,3-dimethyl-2,9-dioxabicyclo[3.3.1]nonane	Attract	Zhao et al. (2019)
<i>Grosmannia europhoides</i>					
<i>G. penicillata</i>					
<i>Ophiostoma bicolor</i>					
<i>O. piceae</i>					
<i>Cryptococcus nemorosus</i>	<i>Spodoptera littoralis</i>		(Z)-hexenyl acetate, ethyl (2E,4Z)-deca-2,4-dienoate (pear ester), (3E)-4,8-dimethylnona-1,3,7-triene (DMNT), linalool, α -terpineol, β -myrcene, (E,E)- α -farnesene, acetoin, ethanol, ethyl acetate, 2-phenylethyl acetate, 2-phenylethanol, (Z)-3-hexenyl acetate, nonanal	Attract	Becher et al. (2018), Ljunggren et al. (2019)
<i>Metschnikowia andauensis</i>					
<i>M. fructicola</i>	<i>Paralobesia viteana</i> ,	Symbiotic			
<i>M. hawaiiensis</i>	<i>Lobesia botrana</i>	Saprophytic			
<i>M. lobburisensis</i>	<i>D. melanogaster</i>	Floral fungi			
<i>M. pulcherrima</i>					
<i>M. saccharicola</i>					
<i>Saccharomyces cerevisiae</i>					
<i>G. penicillate</i>	<i>Ips typographus</i>	Symbiotic	2-Methyl-3-buten-2-ol	Attract	Zhao et al. (2015)
<i>G. europhoides</i>					
<i>Aspergillus flavus</i>	<i>Amyelois transitella</i>	Saprophytic Pathogenic	Conophtherin	Attract	Ampt et al. (2016) Ljunggren et al. (2019)
<i>F. solani</i>	<i>Xylosandrus compactus</i>	Saprophytic Pathogenic	Ethanol, methyl, isovalerate 2,3-butanediol	Attract	Egonyu & Torto (2018)
<i>Penicillium sp.</i>	<i>D. melanogaster</i>	Pathogenic	Geosmin	Repel	Stensmyr et al. (2012)
<i>Saccharomyces cerevisiae</i>	<i>Drosophila</i>	Saprophytic	Lsoamyl acetate, ethyl acetate, ethyl phenyl, acetate	Attract	Christiaens et al. (2014)
<i>F. solani</i>	<i>Diaprepes abbreviatus</i>	Saprophytic	1-Octen-3-ol	Repel	Wu & Duncan (2020)

Continued

Table 1 Continued

Fungi	Insect	Relationship	Chemical	Function	Research
<i>Muscodor vitigenus</i> /	<i>Cephus cinctus</i> <i>Aedes aegypti</i>	Endophytic /	Naphthalene Geosmin	Repel Attract	Daisy et al. (2002) Stensmyr et al. (2012)
<i>Beauveria bassiana</i> <i>Metarhizium robertsii</i> <i>Gliomastix masseei</i>	<i>Cosmopolites sordidus</i> <i>Rhopalosiphum padi</i> <i>Myzus persicae</i>	Pathogenic Saprophytic Saprophytic	3-Cyclohepten-1-one (C ₇), 1,3-dimethoxy-benzene, styrene (C ₁), Benzothiazole(C ₂) Methyl hexadecanoate, methyl 9-octadecenoate, methyl 9,12-octadecadienoate Hexahydroacremoinintriol, acremoinin A, glucoside, hexahydro acremoinintriol	Repel, mask, attract Settling inhibition Settling inhibition	Lozano-Soria et al. (2020) Ruiz-Jimenez et al. (2017) Ruiz-Jimenez et al. (2019)
<i>Acremonium masseei</i>	<i>M. persicae</i> <i>R. padi</i>	Saprophytic	Viomellein, ochratoxin A, brevianamide A citrinin, penicillic acid	Feeding inhibition	Paterson et al. (1987)
<i>Penicillium</i> sp.	<i>D. melanogaster</i> <i>S. littoralis</i>	Pathogenic Saprophytic	Destruixins, cytochalasin D, beauvericin, oosporein, bassianolide, cordycepin, serinocyclin A, beauverolides	Insecticidal virulence	Wei et al. (2017) Molnar et al. (2010)
<i>B. bassiana</i> <i>M. anisopliae</i>	<i>Anopheles stephensi</i>	Endophytic	Nitrosoamide Ergovaline	Insecticidal Repel	Strobel et al. (2010) Gamer et al. (1993)
<i>Muscodor</i> sp. <i>Acremonium</i> <i>coenophialum</i>	/ /	Endophytic Endophytic	Zearalenone	Increase egg production High mortality	Teotor-Barsch & Roberts (1983) Teotor-Barsch & Roberts (1983)
<i>Fusarium</i> sp.	<i>Tribolium confusum</i>	Pathogenic	Trichothothecenes	Retard growth	
<i>Fusarium</i> sp.	<i>Reticulitermes lucifugus</i> <i>Tenebrio molitor</i> <i>T. confusum</i> <i>Chilo zonellus</i> <i>Calliphora erythrocephala</i> <i>Scolytus scolytus</i> <i>Homarus americanus</i>	Pathogenic			
b. Produced by plants stimulated by fungal colonization					
<i>Rhizophagus</i> <i>intraradices</i> <i>B. bassiana</i>	<i>S. exigua</i>	Mycorrhizal EIPF	Myrcene, δ -2-carene, α -phellandrene, sabinene, δ - elemene, β -elemene, (E)- β -caryophyllene, α - humulene /	Repel, decrease fitness	Shrivastava et al. (2015)
<i>Glomus mosseae</i> <i>Glomus</i> sp. <i>Rhizophagus</i> sp. <i>Gigaspora</i> sp. <i>Paraglomus</i> sp. <i>Epichloë uncinata</i>	<i>Aphidius ervi</i> <i>Acyrtosiphon pisum</i>	Mycorrhizal Mycorrhizal	(E)-Caryophyllene, (E)- β -farnesene	Decrease fitness Attract	Guerrieri et al. (2004) Babikova et al. (2014a)
	<i>Teleogryllus commodus</i> <i>Lepidogryllus</i> sp.	Endophytic	Lolines	Repel	Barker et al. (2015)

Continued

Table 1 Continued

Fungi	Insect	Relationship	Chemical	Function	Research
Acremonium sp.	Rhopalosiphum pad Schizaphis graminum Oncopeltus fasciatus	/	N-formyllooline	Insecticidal, repel	Bush et al. (1993)
Melampsora larici- populina	Lymantria dispar	Pathogenic	Mannitol	Promote feeding	Eberl et al. (2020)
c. Produced by fungi as defense compounds against insect predation					
Fusarium sp.	Folsomia candida	Pathogenic	Ausorufain, viomellein, xanthomegnin	Repel	Xu et al. (2019)
Aspergillus sp.	Tenebrio molitor				
Penicillium sp.	Drosophila melanogaster	Saprophytic	Sterigmatocystin, emericellamide, austinol	Repel, reduce fitness	Doll et al. (2013)
A. nidulans	F. candida	Saprophytic	dehydroaustinol	Repel	Zhao et al. (2017)
Neurospora crassa	Sinella curviseta	Saprophytic	Furocoumarin (neurosporin A)		
d. Produced by plants with stimulation of insect behavior, with the presence of fungi					
/	Frankliniella occidentalis,	Mycorrhizal	Myrcene	Repel	Broughton & Harrison (2012)
/	Thrips imagines, T. tabaci				
G. intraradices	Aphidophagous hoverfly	Mycorrhizal	Myrcene	Attract	Stokl et al. (2011)
	Parasitoids	Mycorrhizal	Methyl salicylate	Attract	Babikova et al. (2014a)
G. intraradices	Acyrtosiphon pisum	Mycorrhizal	Methyl salicylate	Repel	Babikova et al. (2014a)
Glomus sp.	A. pisum	Mycorrhizal	(Z)-3-hexenyl acetate, naphthalene, (R)-germacrene D	Repel herbivory, attract natural enemy	Babikova et al. (2014a)
Rhizophagus sp.					
Gigaspora sp.					
Paraglomus sp.					
/, no data.					

enters the pool of dead organic matter (Cebrian, 1999). Fungi and insects can substantially affect decomposition and the carbon/nitrogen recycling rate by degrading and taking up nutrients from plant litter (Tortella et al., 2008; da Costa et al., 2019; Piazza et al., 2021). Specifically, fungal odor intensifies during sporulation to attract highly diverse fungivores, including specialists (plant litter-inhabiting insects) and generalists (Fäldt et al., 1999). As detection signals indicating the presence of decaying fungi and organic material, insects function as decomposers and dispersal vectors are attracted (Fäldt et al., 1999; Jacobsen et al., 2017; Suetsugu et al., 2019). In addition, insects influence the decay process by introducing their associated fungi to dead plant material, then they compete or cooperate with decaying fungi in the plant litter (Jacobsen et al., 2018; Seibold et al., 2022). Furthermore, insect grazers have different food or oviposition preferences and grazing intensities; hence, their presence modifies the fungal community composition. Such behavioral traits determine the activities of some saprotrophic fungal species in the soil and regulate the rates of nutrient turnover in plant organic matter (Newell, 1984; Crowther et al., 2011; Schmalchli et al., 2011). Due to these characteristics, saprotrophic fungi and insects play crucial roles in various types of natural ecosystems (Schmalchli et al., 2011).

4.2 Assist in attracting pollinators

Plant-associated fungi use various strategies to attract pollinators. In some circumstances, the volatile organic compounds released by fungi are true signatures, with functions similar to those of floral scents that attract insect pollinators (Rering et al., 2018). The volatiles from yeast and flowers are widely overlapped and influence pollinator preference, implying the significant role of plant-associated fungi in pollination (Stokl et al., 2010; Ljunggren et al., 2019). Fungi can possibly utilize floral nutrition and produce odors that alter inflorescence odors, thereby attracting insect pollinators (Peach et al., 2021). Insects can be attracted to plants that mimic the scent and appearance of fungi, such as mushroom-mimicking flowers and scents similar to those of mushroom-fruiting bodies, thereby dispersing plant pollen (Kaiser, 2006). Moreover, mycorrhization also increases the visitation rates of pollinating insects by enhancing plant biomass, flower number, and pollen number—these indirect effects should also be considered (Gange & Smith, 2005; Jung et al., 2012).

4.3 Improve plant resistance to insects

By using various strategies, fungi assist plants in defending against herbivores. The colonization of beneficial fungi, such as endophytic and mycorrhizal fungi, improves plant growth by providing nutrients to plants and enhancing plant stress tolerance, which might simultaneously improve their anti-herbivore abilities (Rivero et al., 2021). In addition, fungal colonization primes the chemical defense process in plants by increasing the levels of terpenoids, alkaloids, fatty acid derivatives, and phenylpropanoid polyamine conjugates in plants; these chemicals are released to deter pests (Schausberger et al., 2012; Shrivastava et al., 2015; Rivero et al., 2021). Furthermore, fungal colonization induces chemical volatiles to attract the natural enemies of insects;

such alarm signals can be transferred to neighboring plants (Guerrieri et al., 2004; Song et al., 2019). Another interesting case involves the direct toxicity or killing of pests by plant-associated fungi. For example, EIPF, such as *Beauveria* sp. and *Metarhizium* sp., directly infect and kill insects by transgressing the cuticle or, in some cases, through ingestion (Branine et al., 2019).

4.4 Exacerbate damage inflicted by insects to plants

Colonization by certain fungi, such as ambrosia, pathogenic, and plant-associated entomopathogenic fungi, aggravates the damage inflicted by insects to plants. The volatiles produced by these fungi or their manipulated host plants serve as selection cues for diet or oviposition sites and are detected by insect visitors, including destructive forest pests worldwide (Kandasamy et al., 2019). The approach and settling of these pests could negatively affect host plants. Moreover, the pathogenic *Melampsora larici-populina* alters the nutritional status of *Populus nigra* by inducing the production of mannitol—a sugar alcohol—and the accumulation of amino acids and B vitamins, causing young gypsy moths (*Lymantria dispar*) to prefer feeding on fungi-infected plants (Eberl et al., 2020). In addition, some fungi are true plant decomposers that assist pests in predigesting plant material and overcoming the toxicity of defense chemicals produced by plants, such as terpenoids and phenolics, leading to plant decline and death (Henriques et al., 2006; De Fine Licht et al., 2013; Davis, 2015; Wadke et al., 2016).

4.5 Expand the affected area of agricultural and forestry crops

In addition to causing direct damage, pests can also destroy plants by spreading harmful fungi. Invertebrates are considered important vectors for fungal dispersal (Lilleskov & Bruns, 2005; Jacobsen et al., 2017). Various types of plant-associated fungi release chemical signals to attract insects and provide them with high-quality nutrition; in exchange, the insects facilitate fungal dispersal and colonization of new plants (Kluth et al., 2002). The spread of pathogenic and gallery-dwelling fungi has detrimental consequences on agricultural and forestry crops. These fungi begin the vector process by synthesizing semiochemicals to aggregate insects. This cross-kingdom communication facilitates fungal dispersal. Similar observations have been reported for blue-staining fungi, *Fusarium solani*, *Raffaelea lauricola*, *Puccinia punctiformis*, and *Geosmithia morbida* (Kluth et al., 2002; Luna et al., 2014; Zhao et al., 2015; Blood et al., 2018; Guillen Sanchez et al., 2021). In agriculture and forestry, communication between plant-associated fungi and insects is predominantly mediated by chemical signals, which contribute to plant diseases and pest outbreaks. A better understanding of these mechanisms would help researchers formulate objectives for applied research and fundamental biology (Morath et al., 2012).

5 Future Perspective

In this review, we comprehensively discussed chemical-mediated communication among plants, insects, and fungi. Notably, each organism participates in interactions involving

a complex blend of chemicals and thrives in diverse ecological niches and nutrient resources. To gain a more accurate and comprehensive understanding of the complex ecological relationships among these organisms, it is crucial to study compound-mediated interactions in their natural habitats. However, as a relatively new research field, several questions remain unaddressed.

First, the functions of bioactive chemicals should be more precisely defined, and the molecular mechanisms operating in each organism should be thoroughly investigated along with the driving force that selects these compounds as functional molecules in nature. In particular, the mechanisms by which bioactivities enhance plant biomass by inducing plant growth, inhibiting fungal pathogens, repelling herbivores, and attracting pollinators should be investigated. They may present defense, competition for nutrients and space, the ability to attract beneficial insects, and the induction of host resistance as their primary and secondary mechanisms of biocontrol activity. Moreover, the metrics by which the presence, abundance, and functions of application of potential bioactive compounds can be measured remain unclear. Future assessments of bioactive compounds would benefit from the combination of electroantennogram techniques, gene editing technology, and metabolomics. In some cases, isolated chemicals are initially considered metabolic waste, but their functions are unclear. However, some microbial molecules have recently been identified as functional chemical signals (Mehmood et al., 2019; Tourneroc et al., 2019). Therefore, understanding the mechanistic and molecular bases of chemical signaling pathways could help identify additional chemical signals and shed light on their roles in mediating communication among organisms. Determining the modes of action of chemicals and their producers is useful for assessing their bioregulatory potential and developing appropriate application methods for the effective management of eco-agriculture.

Second, it is important to investigate whether bioactive chemicals and metabolites with core structures and their analogs exert similar functions as well as ecological importance, as the key active molecules produced by different organisms are relatively conserved and their functions may have converged during evolution, such as terpenoids and bis-naphthopyrones (Schiestl, 2010; Zhou, 2018; Xu et al., 2019). This requires an understanding of the organization, regulation, and diversification of core chemical metabolic pathways and addresses the important functions of metabolites during interactions. Clarifying whether natural products and their analogs exert similar functions during plant–fungi–insect interactions is crucial for the application of bioactive molecules. This knowledge will enhance our understanding of these interactions and the potential use of bioactive compounds as environment-friendly strategies for artificial ecology management.

Third, the interactions between different bioactive compounds, such as repellents or attractants, require further investigation. Currently, most studies focus on the effects of a single compound. However, communication between plant-associated fungi and insects is usually achieved using a combination of chemicals. In most cases, the functions of different compounds may offset, reinforce, or intersect with each other. Therefore, investigating individual compounds

may not fully reflect the actual scenario (Weisskopf et al., 2021). For example, a volatile repellent can augment the function of known repellent compounds, suggesting a cost-efficient way of protecting the host against insects (Unelius et al., 2014).

Finally, it should be noted that, although many studies have been carried out in experimental settings, the complexity of chemical communication under natural conditions must also be considered. Investigations of the chemical signals that modulate interactions should be undertaken within the context of a complex ecological system that includes variations in temperature, soil type, C: N ratios, and humidity (Noman et al., 2020). With the discovery of more bioactive molecules, it is important to understand the effects of these chemical signals on the adaptability of organisms to changing environments.

6 Concluding Remarks

In this review, we comprehensively summarized the interaction patterns between plant-associated fungi and insects, and discuss their ecological significance and future prospects. The observation that chemical molecules from plants and fungi promote plant growth and influence insect activity indicates their potential applications as plant growth promoters and insect behavior regulators. Overall, bioactive compounds and their mechanisms of action in modulating interactions across kingdoms have not been well studied, indicating a research area ripe with opportunities for interesting discoveries. Furthermore, detailed studies on these interactions are required to facilitate the identification and screening of active compounds or beneficial plant-associated fungi as part of an agroecosystem with balanced efficiency and sustainability.

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Conflict of Interest

The authors declare no competing financial interests.

Authors' Contributions

H.H. had the idea for the article, Y.X. performed the literature search and wrote the draft of the manuscript, H.H. and L.X. critically revised the work.

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